Description for programs ODFR4_3 and ODFRM4_3.

Introduction.

The software ODFR is intended for simulation of slow motional EPR spectra for paramagnetic probes, first of all, for nitroxide radicals in the partially oriented samples. The main purpose of the program is the joint simulation of EPR spectra, recorded for the anisotropic medium at different sample orientation in the spectrometer magnetic field (EPR spectrum anguluar dependence). In the program it is assumed that EPR spectrum anguluar dependence was obtained by turning of the sample around one axis on different angles. The program **ODFR4_3** is single-processor version. The program **ODFRM4_3** is intended for multiprocessor MPI execution. The number of MPI processes should be taken as multiple of number NORT (number of fundamental orientations for calculations in the framework of ODF-model, see magnetic parameter file description, Line 10). The detailed description of all models and approximations used in the program is contained in the following reference:

A.Kh.Vorobiev, A.V.Bogdanov, T.S.Yankova, N.A.Chumakova, "*Spin probe determination of molecular orientation distribution and rotational mobility in liquid crystals. Model-free approach.*", submitted to **Phys. Chem. Chem. Phys.**, 2018.

Please, cite this reference whenever you intend to publish results obtained with the software ODFR. If you have any questions about the program usage or maintenance, please contact A. Kh. Vorobiev at: a.kh.vorobiev@gmail.com

Kernel of the ODFR program is taken from program NLSL, Version 1.5.1b, described in [1]. The main differences are the following:

1) The program ODFR4 explicitly uses an orientation distribution of the local axial director in the sample frame. This distribution is described by expansion in a series of elements of Wigner D-matrix.
2) High-frequency molecular reorientations described by quasi-libration model are incorporated.
3) Additional angular-dependent hyperfine splitting can be considered. The quantum numbers for the additional nuclei are not included in the basis set for SLE simulation.
4) Interface of the program is changed completely.
5) Some bugs in the program NLSL were improved (see list of improvements presented below.)

Reference:

[1] D. E. BUDIL, S. LEE, S. SAXENA, J. H. FREED, "Nonlinear-Least-Squares Analysis of Slow-Motion EPR Spectra in One and Two Dimensions Using a Modified Levenberg–Marquardt Algorithm", JOURNAL OF MAGNETIC RESONANCE, Series A 120, 155–189 (1996)

Program installation and test tasks.

The program is written in FORTRAN language consistent with FORTRAN77 standard. The source files are provided, as well as the compiled execution file ("ODFR4_3.exe"). No additional libraries are needed for compilation of the program. The performance of the program was tested in Windows and Linux operating systems, Intel Fortran compiler was used.

To launch the program, put the executable file in the directory with parameter files, and start the executable file. The names of parameter and EPR spectra files are listed in file "nameODFR.nam" (see description below).

To check the program performance, some example sets of parameter files are included in the distribution. These examples are contained in the file "ODFR_Tests.zip". The examples are listed below:

- the folders "sampl1" - "sampl5" present the examples suggested in [1] for program NLSL, Version 1.5.1b.
- folder k18p-1d contains the result of fitting using mean field potential model (MF) in the case of angular dependence of EPR spectra for nitroxide probe A5 in SmA liquid crystal H59. This example includes quasi-librations and additional components resulting from natural abundance of ^{13}C-isotope. Only 8 out of 16 simulated spectra are presented for diminishing the calculation time.
- folder k18-6c contains the final result of simulation of the same spectra in the framework of orientation distribution function (ODF) model.
- folder F322-21br presents the similar results for A5 in axial nematic 8CB liquid crystal.
- directory F16nr illustrates the case of small orthorhombic distortion of axial orientation distribution function in the case of the probe A5 in HOPDOB (SmA) liquid crystal.
- folder F93-7r demonstrates the case of non-orthorhombic orientation distribution.

The general description of the input files.

All options and parameters used for control of the program operation are contained in parameter files. These files contain parameters of calculation (integer and real numbers) as well as keys and flags (integer numbers).

Most of the calculation parameters can be varied to minimize the discrepancy between the experimental spectra and the simulated ones. The variation of these parameters is controlled by variation flags provided in the parameter files. The flags indicate to the program, whether each parameter should be kept fixed, or varied, or comply with some symmetry (axial or isotropic tensors), or keep the fixed proportional ratio with some other parameter. The detailed description of possible variation flags will be given below.

Each simulated **experimental spectrum** should be provided in a separate **file**. The file should contain two columns of data, the first one with the magnetic field values, and the second one with the spectrum intensity values. The data should be preceded with the line containing the keyword "`Spectr:`". Before this keyword, any additional information may be listed (conditions of spectrum recording, experiment description, etc.). This additional information is not interpreted by the program.

Each simulated spectrum is supplied with the corresponding **spectral parameter file** (extension "`*.par`"). The files differ by the value of angle `psi` describing the orientation of the sample with respect to the spectrometer magnetic field. In most of the calculations, all other parameters are the same for all simulated spectra, since the spectra correspond to the different orientations of *the same sample*. The first spectral parameter file is a master-file. The parameters marked as varied in this file will change in all other parameter files simultaneously. However, there is a possibility to vary some parameters independently for the desired spectra. This situation may be of use to independently vary the amplitude for each spectrum, in order to compensate the errors of the spectra normalization; or to accurately determine the angular dependence of line widths etc.

It should be noted that the number of parameters independently varied in each parameter file should be explicitly stated (see Line 2 of magnetic parameter file).

Note that each spectrum can contain contributions from several types of paramagnetic species contained within the sample. These different species (or "paramagnetic sites") may correspond to:

- several chemically inequivalent radicals contained in the same sample;
- several types of localization of chemically identical probes, differing in magnetic parameters, or rotation mobility, or linewidth or so on.
- contributions from isotopomers, corresponding to natural abundance of magnetic isotopes of ^{15}N, ^{13}C, $^{35/37}$Cl etc., or from isotopically labeled radicals etc. etc.

In the case of several paramagnetic sites contained in the sample, the parameters for every site are listed in each spectral parameter file. It should be noted that the easiest way to introduce the additional site in the set of parameter files is the following: the additional site is described in the master-file and marked as dependent by assignment value 1 to the key `ItypeSite` (see detailed description below). In this case additional site will appear in all other parameter files. The parameters for additional sites can be fixed, or varied independently, or constrained according to the corresponding parameter of the first site. This behaviour is controlled by a flexible system of variation flags, which will be described in detail below

The orientation distribution of the probe molecules is controlled by parameters contained in **orientation distribution function (ODF) file**. Note that each paramagnetic site can possess a separate ODF, or all ODFs can be fixed as equivalent to that of the first site. The corresponding key value (`keyODF`) is described in detail below.

Finally, the names of all files used in the simulation are listed in **the name file** (its own name is fixed and is always "`namODFR.nam`"). The name file contains the filenames of

1) the ODF file (input and output);
2) all the experimental spectra files;

3) the corresponding spectral parameter files (input and output);
4) the type of ODF used for each simulated spectrum.

The name file can be constructed in such a way as to perform up to 10 sequential tasks one after the other. This feature can be useful, for example, to perform the spectra simulation in iterative manner, with consecutive approximations, or to simulate a series of spectra with related parameters, e.g. temperature, pressure, concentration dependences of spectra etc. Note that all the output files of the program ODFR can be recycled as input files for further calculations.

The detailed description of the structure of input/output files

Name file "namODFR.nam" contans the names of files used in calculation. Sequential performance of up to 10 tasks is possible (`numSim`).

Line 1 for any task is the arbitrary integer number. It is not used by program.

Line 2 in the description of task contains:

- name of file describing initial orientation macroscopic distribution function of local director (below, ODF), name of file describing result ODF and number of simulated spectra (`NSpec`) separated by comma; `NSpec` should not be more `MXSPC`=20.

Line 3 and furthers describes each one of the simulated spectra.

Each line contains:

- file name of initial parameters for spectrum, file name of result parameters, file name of experimental spectrum, file name of simulated spectrum and integer key, designating ODF type (`ODFtype`). All are separated by comma.

`ODFtype=0` – absence of ODF (isotropic spectra or MF model);
`ODFtype=1` – macroscopic ODF, described by potential (dlam);
`ODFtype=2` - orthorhombic ODF, described by even order parameters;
`ODFtype=3` – non-orthorhombic ODF, described by even and odd order parameters.

List of tasks is finished by a line containing keyword "end".

Input ODF file (see ODFR4F.par in the source file directory as the example):

Line 1 - integer key `ODFcontrol`.

`ODFcontrol` means:.

`ODFcontrol=0` - using the presented ODF with variation of parameters marked by nonzero flags;
`ODFcontrol=1` - using the axial ODF, non-axial parameters become zero;
`ODFcontrol=2` - using frozen ODF, parameters are not varied regardless of flags;
`ODFcontrol=3` - transformation of orthorhombic ODF to non-orthorhombic one; the odd order parameters are added in the ODF file; variation flags for new parameters are assigned zero; `ODFcontrol` after procedure becomes zero as well;
`ODFcontrol=4` - transformation of non-orthorhombic ODF to orthorhombic one; the odd order parameters are removed from the ODF file; `ODFcontrol` after procedure becomes zero.

Line 2 contains:

- key for penalty for negative part of ODF, penalty coefficient and number of integration nodes for calculation of negative part, separated by spaces.

When the complex spectrum with two or more sites (paramagnetic species) is calculated, ODF for each site should be described separately. The parameters of orientation distribution can be varied independently, or can be kept the same as in the first site. To control this, the flag `keyODF` is used (see below, Line 4). Description of ODF for the second site follows the description of ODF for the first site (starting with Line 3). Please note that calculation of the spectra for 3 or more sites with different ODFs was not tested. The total number of paramagnetic sites (`nnsite`) is defined in the magnetic parameter file, Line 2. See Test parameters in folder k18p-1d for example.

Line 3 is not used.

Line 4 contains:

- integer even number is rank of ODF (Jmx) and key describing the type of the ODF for this site/probe (keyODF), separated by spaces.
keyODF = 0 - independent ODF;
keyODF > 0 - ODF is identical to first ODF;
keyODF <0 - isotropic ODF (MOMD) for this site/probe.
Line 5 is not used.
Line 6 and further contains real values of ODF expansion with increasing indices. Each coefficient is accompanied by integer flag (iflagODF), which indicates varied (iflagODF=1) or unvaried (iflagODF=0) the parameter. The last line corresponds to coefficients with rank Jmx.
If the number of paramagnetic sites is more than 1, then the description of ODF for the second (and so on) site should follow, starting with Line 3.
After description of all sites, all the remaining lines are ignored.

Output ODF file describing resulting ODF contains additionally:
- fraction of the negative part of result ODF (line 3);
- order parameter for macroscopic ODF Pjm0;
- order parameters for local distribution, described by potential Pj0n;
- global order parameters, calculated as Pjmn=(Pjm0)*(Pj0n)
Output file can be used as input file for the next simulation, additional lines are ignored.

Input file of spectral parameters (see ODFR4.par in the source file directory as the example):
The first file of the spectral parameters is the main (master-file). If it is not specially designated, many values from this file are used for other spectra simulated.
Line 1: kcontrol keyiparm keyCov separated by spaces
- kcontrol operates minimization procedure:
kcontrol = 0 - minimization using MINPACK procedure as in original program;
kcontrol = 1 - stop calculation, for termination of program execution the value 1 should be written in the place of this value in file "temp 1.par";
kcontrol = 2 - single calculation of spectra without minimization by variation of parameters;
kcontrol = 3 - minimization using NL2SOL procedure (recommended).

- keyiparm allows using for calculation of the spectrum the values differing from values presented in master-file:
keyiparm = 0 - parameters as in first file (master-file);
keyiparm = 1 - independent parameters;

- keyCov defines the procedure for calculation of covariance matrix (for procedure NL2sol):
keyCov = 0 - covariance matrix is not calculated;
keyCov = 1 - covariance matrix calculated using Hessian and Jacobian,
scale$\cdot H^{-1}\cdot (J^{T}\cdot J)\cdot H^{-1}$
keyCov = 2 - covariance matrix calculated using Hessian,
scale$\cdot H^{-1}$
keyCov = 3 - covariance matrix calculated using Jacobian,
scale$\cdot (J^{T}\cdot J)^{-1}$

Line 2 contains:
- number of varied parameters, designated **in this file** (filenprm),
- number of paramagnetic centers, or sites, described in this file (nnsite < 5)

Line 3 contains:
- number of calculated points in the spectrum,

- number points used for linear spline of experimental points at treatment of experimental spectrum.

Line 4 contains:
- coefficient defining the direction for experimental spectrum,
- coefficient defining the direction for result spectra.

Line 5 contains:
- shift of experimental spectrum on magnetic field (ShiftExp) in Gauss,
- noise level (rmsn),
- key (iwflag) turning weighted residual fit (using MINPACK):
`iwflag = 0` - turned off
`iwflag <> 0` - turned on.
Using the noise and weighted residual fit were not tested.

Line 6 contains:
- centrum of calculated spectrum,
- range to the lower field,
- range to the higher field.
All values are in Gauss.

Line 7 contains:
- tolerance for finishing of minimization by parameters vector (XTOL),
- tolerance for finishing of minimization by function value (FTOL),
- tolerance for finishing of minimization by gradient value (GTOL).

Line 8 contains:
- doubled nucleus spin (in2),
- nuclear gyromagnetic ratio (gamman (radians / sec*gauss))

Line 9 contains:
- number of additional nuclei hfi,
- their doubled spins.

Line 10 contains:
- number of orientation (`NORT`) for calculation of MOMD and for any non-isotropic macroscopic ODF,
- `KEYLIB` defines type of quasi-librations:
`KEYLIB = 0` without quasi-librations,
`KEYLIB = 1` small-angle quasi-librations (sin2),
`KEYLIB = 2` quasi-librations in accordance with [Van SP, Birrell GB, Griffith OH. J. Magn. Reson., 1974;15: 444],
`KEYLIB =` quasi-librations in accordance with [Timofeev V, Samarinov B (1995). J Chem Soc Perkin Trans. 2: 2175–2181],
`KEYLIB = 4` approximation of 3D quasi-librations in accordance with [A.Kh.Vorobiev, N.A.Chumakova in "Nitroxides - Theory, Experiment and Applications", Ed. A.I.Kokorin, 2012, p58-112].

Line 11 contains:
- key defining the rotation model (`ipdf`),
- number of terms in potential (`ipt`).

Brownian rotational diffusion (`ipdf=0`) only was tested as it is the sole model consistent with non-isotropic ODF.

Line 12 contains:
- key (itw) defines coordinate frame for Lorentzian line width,
itw = 0 - g-tensor frame,
itw = 1 - A-tensor frame,
itw = 2 - D-tensor frame.
- number of positions in jump rotation model (was not tested)

Line 13 contains:
- tolerance (`cgtol`) for procedure of tridiagonalization,
- number of steps (nstep) for procedure of tridiagonalization (nstep=0 means unrestricted).

Line 14 contains:
- truncation values [lemx,lomx,kmx,mmx,ipnmx]

Line 15 contains:
-keys for non-Brownian rotations ml,mxy,mzz, (was not tested).

Line 16 - is not used.

Lines 17 and furthers, each line contains three parameters, which can be varied. Fitting is managed by flags after the parameter.
Content of these lines sequentially:
Spectrum shift, microwave phase, content factor
Gauss line width (gib0), (gib2), (gibX), (rsrv)
if gibX=0 then Gauss width = $(gib0+gib2\cdot\sin^2(psi))$;
else Gauss width = $(gib0\cdot\cos^2(psi)+gibX\cdot\sin^2(psi))$;
(both, gib2) and (gibX, gib2) are valid for MOMD and `ipt>0` only;
Lorentz line width (wxx,wyy,wzz),(rsrv)
g-values (gxx), (gyy), (gzz)
A-values (Axx), (Ayy), (Azz)
magnetic tilt (alm), (bem), (gam) degrees
diffusion tensor (Dx), (Dy), (Dz) 1/sec
diffusion tilt (ald), (bed), (gad) degrees
[pml,pmxy,pmzz] were not tested
discrete jumps freq. (djf),(djfprp), Heisenberg spin exchange rate (oss) s^{-1}
potentials (c20), (c22), (c40)
potentials (c42), (c44), macroscopic potential (dlam=dc20)
libration amplitudes (Lx), (Ly), (Lz) degrees
libration tilt (alphaL), (betaL), (gammaL)
resonantor frame (rsrv,psi,rsrv) - changing the angle psi means the rotation around Y axes of the sample; variation of angle psi in the program simultaneously changes all values psi for all spectra.

Lines describing additional hfi:
Parallel constants of hfi, perpendicular constants of hfi, key of equivalence:
(Apar) (flag) (Aperp) (flag) (iEQs)
Additional hyperfine splitting is assumed axial. Principal axis Apar corresponds to sample orienation psi=0.
Key of equivalence iEQs is number of nucleus equivalent to this one.

If the spectrum is the sum of few spectra (not more then 5) next line should contain key (ItypeSite)
ItypeSite = 0 - independent paramagnetic center (site),
ItypeSite > 0 - paramagnetic center (site) identical to the first one but has additional hfi; in this case the varied parameters for this site can be only constants of hfi, Gauss line width and content factor.

Second and further paramagnetic sites are described by similar lines starting from line 8
Parameters marked (rsrv) do not used.
Following flags can be used for managing the fitting:

0 - invariable parameter;
1 - varied parameter;
2 - varied parameter is transformed as follows:
in the case components of g-tensor as (g-2.0)*1000;
in the case diffusion coefficients log(D);
3 - value is calculated from other varied parameter; if one component of g-tensor is marked by flag 3, it will be calculated from others with retaining of g averaged
4 - axial (mark valY, means valY=valX)
5 - isotropic (mark valY and valZ, means valY=valX and valZ=valX)
6 - mark the parameter for all spectra as independent variables
7 - mark the parameter as variable equal for all spectra together
8 - mark the parameter as invariable but independent for all spectra

10 - parameter for spectrum is independent but is not varied
11 - parameter for the spectrum is varied independently from the same parameter for other spectra
12 - independently vary as transformed
13 - calculated, 14 - axial, 15 -isotropic

91 - parameter = the same parameter of 1st site with initial ratio
92 - parameter = the same parameter of 2nd site with initial ratio
93 - parameter = the same parameter of 3rd site with initial ratio

Euler angles, first method:
- rotation around z axis - alpha (0-2Pi);
- rotation around new axis y1 - beta (0-Pi);
- rotation around new axis z2 - gamma (0-2Pi);

Euler angles, second method, the same transformation:
- rotation around z axis - gamma (0-2Pi);
- rotation around old axis y - beta (0-Pi);
- rotation around old axis z - alpha (0-2Pi);

Euler angles in program transform the specified frame (for example, rotation frame) to magnetic frame (g-tensor frame). To use the opposite transformation, angles (-gamma, -beta, -alpha) should be used.
It is equivalent to the set (Pi-gamma, beta, -Pi-alpha).
Values fi, psi, ksi= 0, angle value, 0 means the rotation around Y axes.

In the course of execution, the program creates following files:
- temp 1.par, temp 2.par,… temp MXSPC.par, - They contain current values of parameters. To terminate calculation, the first position in the file temp 1.par should be changed to 1;
- temp 1.dat, temp 2.dat,… temp MXSPC.dat, - Current spectra are in the files.

- file 'history' - contains the copy of screen output (it is logout file). This file are not rewritten, but it is filled in an `'APPEND'` mode. The covariance matrix is written in this file.
After finishing task, the program creates files of result ODF, result parameters for each spectrum and result spectra. The file for every result spectrum contains the experimental spectra and calculated ones as columns. The spectra for every site are presented separately in the columns after the sum spectrum. The magnetic parameters averaged by quasi-librations are presented in the beginning of the spectra files.

List of improvements in the program kernel:

1) in file lcheck.f

The fragment:

```
      c     --- Get truncation value for antisymmetric K-space if needed
            if (jkmn.eq.-1) then
               kmn=-kmx
            else
              kmn=0
            endif
```

is omitted in the NLSL in comparison with previous version of program.

The result is positive or zero kmn in the cases of alpha or gamma magnetic or diffusion tilt angles.

2) in the file lcheck.f

The fragment:

```
      c     Get truncation value for antisymmetric M-space if needed
            if (jmmn.eq.-1) then
               mmn=-mmx
            else
               mmn=0
            endif
```

is omitted in the NLSL in comparison with previous version of program.

The result is incorrect M in the case of nonzero nuclear Zeeman interaction

3) in file matrll.f

The fragment:

```
                  if((ksabs.le.2).and.fld) then
                     if (jkd.eq.0) then
                        z=w3j(lr,2,lc,kr,-ks,kc)
                        ra2=fad(1,ks+3)*z
```

The result is incorrect calculation ra2, rg2, rd2.

The fragment should be changed to:

```
                  if((ksabs.le.2).and.fld) then
                     z=w3j(lr,2,lc,kr,-ks,kc)
                     if (jkd.eq.0) then
                        ra2=fad(1,ks+3)*z
```

4) in file matrll.f

The fragment:

```
                  kcmx=min(lc,kmx)
                  if (nrow.eq.ncol) then
                    kcmn=kr
                  else
```

The result is the nonnegative kcmn and loss of some members.

The fragment should be changed to:

```
                  kcmx=min(lc,kmx)
                  if (nrow.eq.ncol) then
                    kcmn= krsgn
                  else
```

5) in file matrll.f

The fragment:

```
                    mcmx=min(lc,mmx)
                    if (nrow.eq.ncol) then
                      mcmn=mr
                    else
```

The result is the nonnegative mcmn and loss of some members.

The fragment should be changed to:

```
mcmx=min(lc,mmx)
if (nrow.eq.ncol) then
  mcmn=mrsgn
else
```

Mistake in PC.zip
1) in file matrll.f
The fragment:

```
      if (ksi.le.li.and. abs(xlk(li,ksi)).ge.RNDOFF)
#         cdffu=cplkc*xlk(li,ksi)*w3j(lr,l,lc,kr,-ks,kc)
```

In accordance with eprll and expression (A40) in J.Chem.Phys. 77, 3915 (1982), the fragment should be changed to

```
      if (ksi.le.li.and. abs(xlk(li,ksi)).ge.RNDOFF)
#         cdffu=cjkc*cplkc*xlk(li,ksi)*
#        w3j(lr,l,lc,kr,-ks,kc)
```